



Classification Data Mining

BSCS 3A

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OUTLINE

- **Classification**
- Types of Classification
 - Binary, Multi-class, Multi- Label
- Classification Techniques
 - Decision Tree based Methods
 - Rule-based Methods Memory based reasoning
 - Neural Networks / Deep Learning
 - Naïve Bayes and Bayesian Belief Networks
 - Support Vector Machines



“The ₱50,000 Mistake”

“You are the branch manager of Jollibee.
Yesterday, you predicted LOW sales... so you scheduled fewer staff and prepared less food.
But something unexpected happened...
It was:

- ✓ Saturday
 - ✓ There was a promo
 - ✓ Weather was good
- Result: **Sales exploded.**

“**What went wrong?**”

Customers:

- ✓ Waited too long
- ✓ Complained
- ✓ Some left

The branch lost **₱50,000 in potential sales.**

“The ₱50,000 Mistake”

Now your boss tells you:

- ✓ This should not happen again.
- ✓ We have years of data—use it.
- ✓ “Your job is to build a system that can answer this question:
Will tomorrow’s sales be HIGH or LOW?”

“This problem is exactly what we call **classification** in Data Mining.”

“Based on your intuition, what factors affect Jollibee sales?”

- ✓ Day (Weekend?)
- ✓ Weather
- ✓ Promo
- ✓ Time

Classification

Classification is a **supervised learning technique** in Data Mining where the goal is to assign data instances into **predefined categories (classes/labels)**.

- ✓ It learns from **labeled training data**:
Input (features) → Output (class label)

Jollibee Sales-Based Definition

At **Jollibee**, classification could answer questions like:

- ✓ Will a promo be **successful or unsuccessful**?
- ✓ Is today's sales **High, Medium, or Low**?
- ✓ Will a customer **buy Chickenjoy or not**?

Classification

Example

Time	Day	Promo	Weather	Sales Level
12PM	Sat	Yes	Sunny	High
3PM	Mon	No	Rainy	Low
6PM	Sun	Yes	Cloudy	High

The model learns:
“If it's weekend + promo → High sales”

Why Classification is Important

Classification helps **decision-making**.

In Jollibee:

- ✓ Predict **peak hours**
- ✓ Decide **staff scheduling**
- ✓ Plan **promotions**
- ✓ Optimize **inventory**

Supervised Learning Concept

Classification is **supervised learning**, meaning:

- The model learns from **labeled data**

Input (features): Time, Promo, Weather

Output (label): Sales Level

Key Idea

$$f(X) \rightarrow Y$$

Where:

- X = features (Time, Promo, Weather)
- Y = class (High/Medium/Low Sales)

How Classification Works (Step

Step 1: Collect Historical Data

Jollibee collects past sales records

Step 2: Learn Patterns

Example:

Weekend + Promo → High Sales

Weekday + No Promo → Low Sales

Step 3: Build Model

Algorithm finds relationships

Step 4: Predict New Case

Example:

Sunday + Promo + Sunny

→ Model predicts: **High Sales**

How Classification Works at Jollibee®



Intuitive Analogy (Jollibee Version)

Think of a **store manager**:

After years of experience:

“Fridays are always busy”

“Rain reduces dine-in sales”

✓ The manager is doing **manual classification**

✓ The algorithm automates this

Classification vs Other Data Mining Tasks

Task	Goal	Jollibee Example
Classification	Predict category	High vs Low Sales
Prediction (Regression)	Predict number	Sales amount
Clustering	Group data	Customer segments

Types of Classification Outputs

WHY “TYPES OF OUTPUT” MATTER

- The **type of classification output depends on the business question**

At Jollibee:

- ✓ “Will sales be high?” → Binary (Yes or No)
- ✓ “What level of sales?” → Multi-class (Low, Medium, High)
- ✓ “What items will the customer buy?” → Multi-label (Spaghetti and Chicken, Chicken and Fries, Burger and Fries)

Same dataset → Different output types depending on the question

Types of Classification Outputs

BINARY CLASSIFICATION - Only **two possible classes**

Promo	Sales $\geq$ ₱50k	Label
Yes	Yes	1
No	No	0

Example 2: Customer Behavior

- ✓ Will customer buy Chickenjoy?
 - ✓ Yes / No

Example 3: Store Operations

- ✓ Will branch be crowded?
 - ✓ Crowded / Not Crowded

Output:
•1 = Successful
•0 = Not Successful

Types of Classification Outputs

MULTI-CLASS CLASSIFICATION - More than two classes

Sales Amount	Class
< ₱20k	Low
₱20k–₱50k	Medium
> ₱50k	High

Example 2: Customer Type

- ✓ Student
- ✓ Office Worker
- ✓ Family

Only **one class per instance**

Example:

- ✓ A day cannot be both “Low” and “High” sales

Example 3: Time-Based Demand

- ✓ Breakfast Rush
- ✓ Lunch Peak
- ✓ Dinner Peak

Types of Classification Outputs

MULTI-LABEL CLASSIFICATION - One instance can belong to multiple classes simultaneously

Example 1: Customer Order Prediction

Customer may buy:

- ✓ Chickenjoy ✓
- ✓ Spaghetti ✓
- ✓ Burger ✗

Output:

- ✓ {Chickenjoy, Spaghetti}

Example 2: Marketing Tags

Customer behavior:

- ✓ Budget-conscious
- ✓ Family-oriented
- ✓ Frequent buyer

A single customer can have **all three**

Example 3: Basket Analysis (Advanced)

Predict:

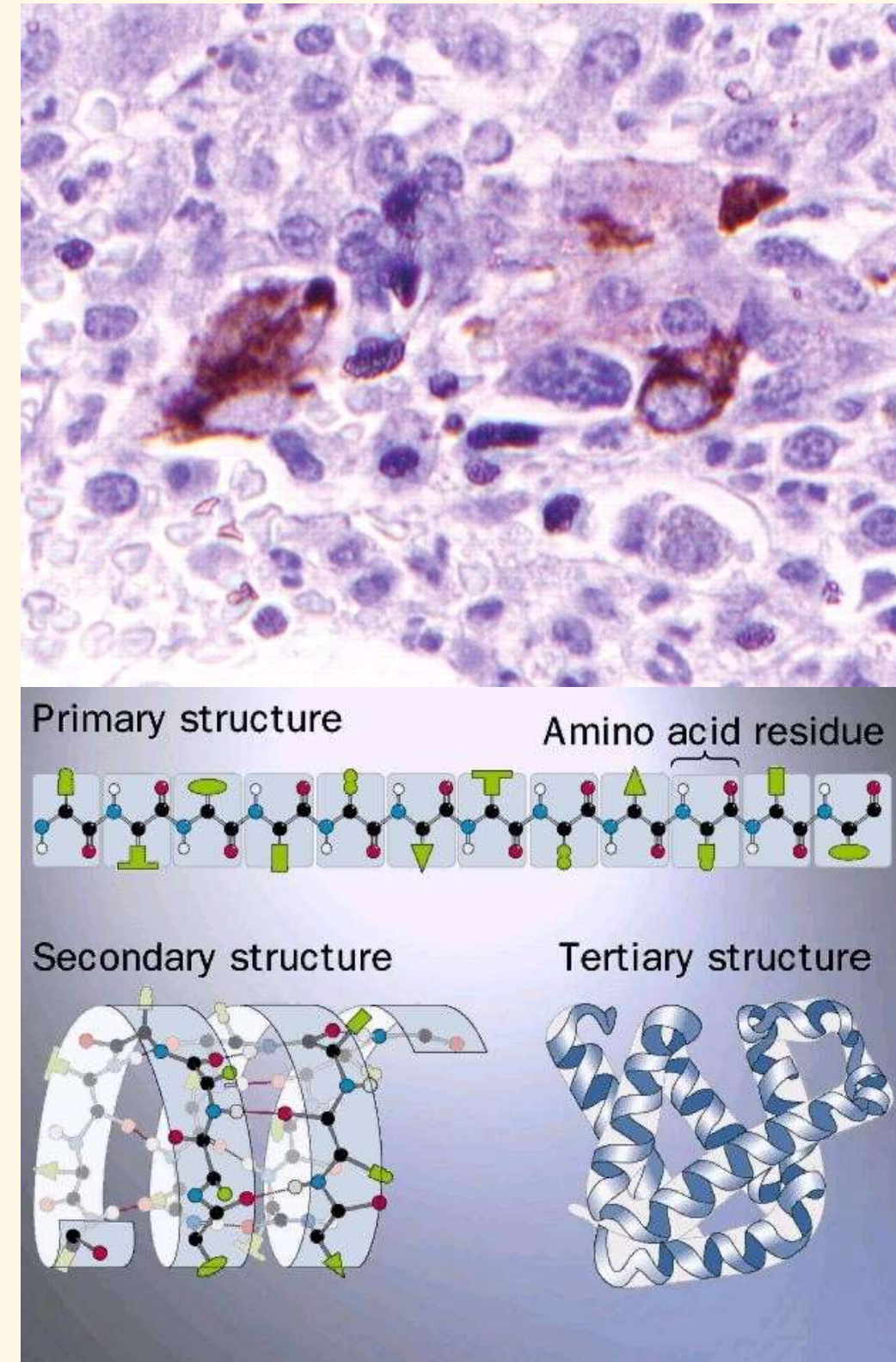
- ✓ Items likely to be purchased together

➤ NOT mutually exclusive

Example: Classifying Movies based on Genres: drama, comedy, science fiction, fantasy, horror, thriller, western, and musical

Examples of Classification Task

- Predicting tumor cells as benign or malignant
- Classifying credit card transactions
 - as legitimate or fraudulent
- Classifying secondary structures of protein
 - as alpha-helix, beta-sheet,
 - or random coil
- Categorizing news stories as finance, weather, entertainment, sports, etc



CLASSIFICATION techniques



Decision Tree based Methods



Rule-based Methods



Memory based reasoning



Neural Networks / Deep Learning



Naïve Bayes and Bayesian Belief Networks



Support Vector Machines

DECISION TREE–BASED METHODS

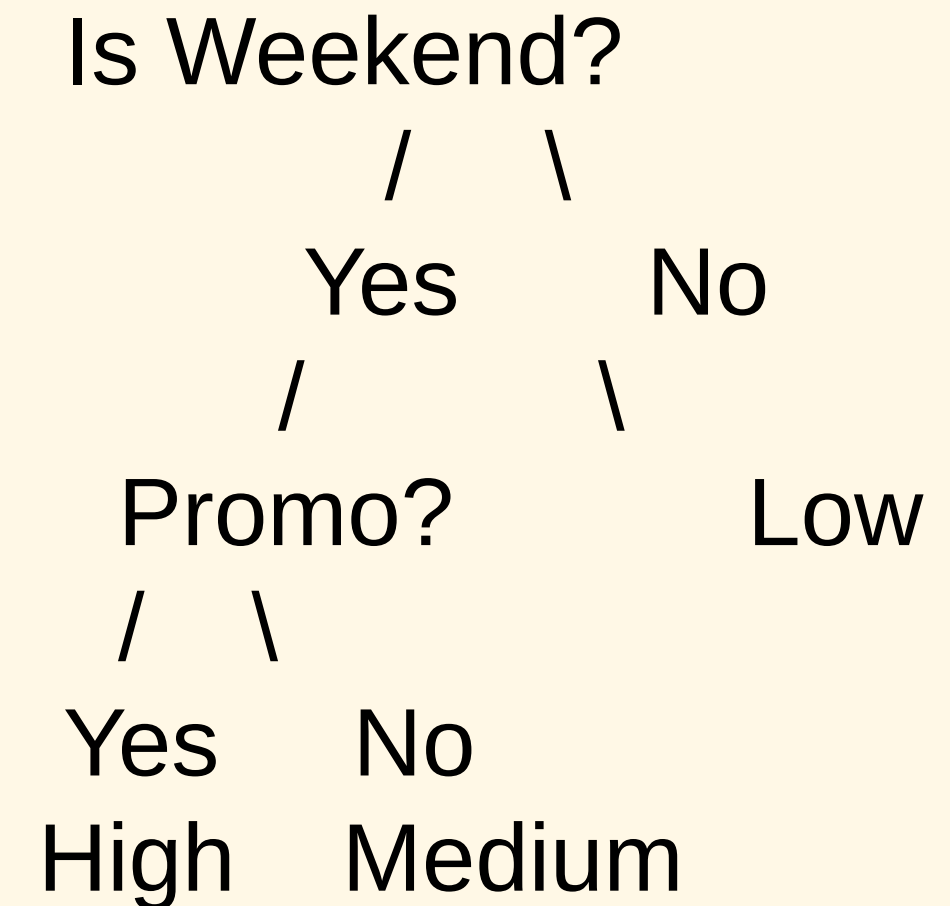
A **Decision Tree** is a tree-like model where:

- ✓ Internal nodes = conditions (features)
- ✓ Branches = decisions
- ✓ Leaves = class labels

Goal: Predict **Sales Level (High/Low)**

Day	Promo	Weather	Sales
Weekend	Yes	Sunny	High
Weekday	No	Rainy	Low

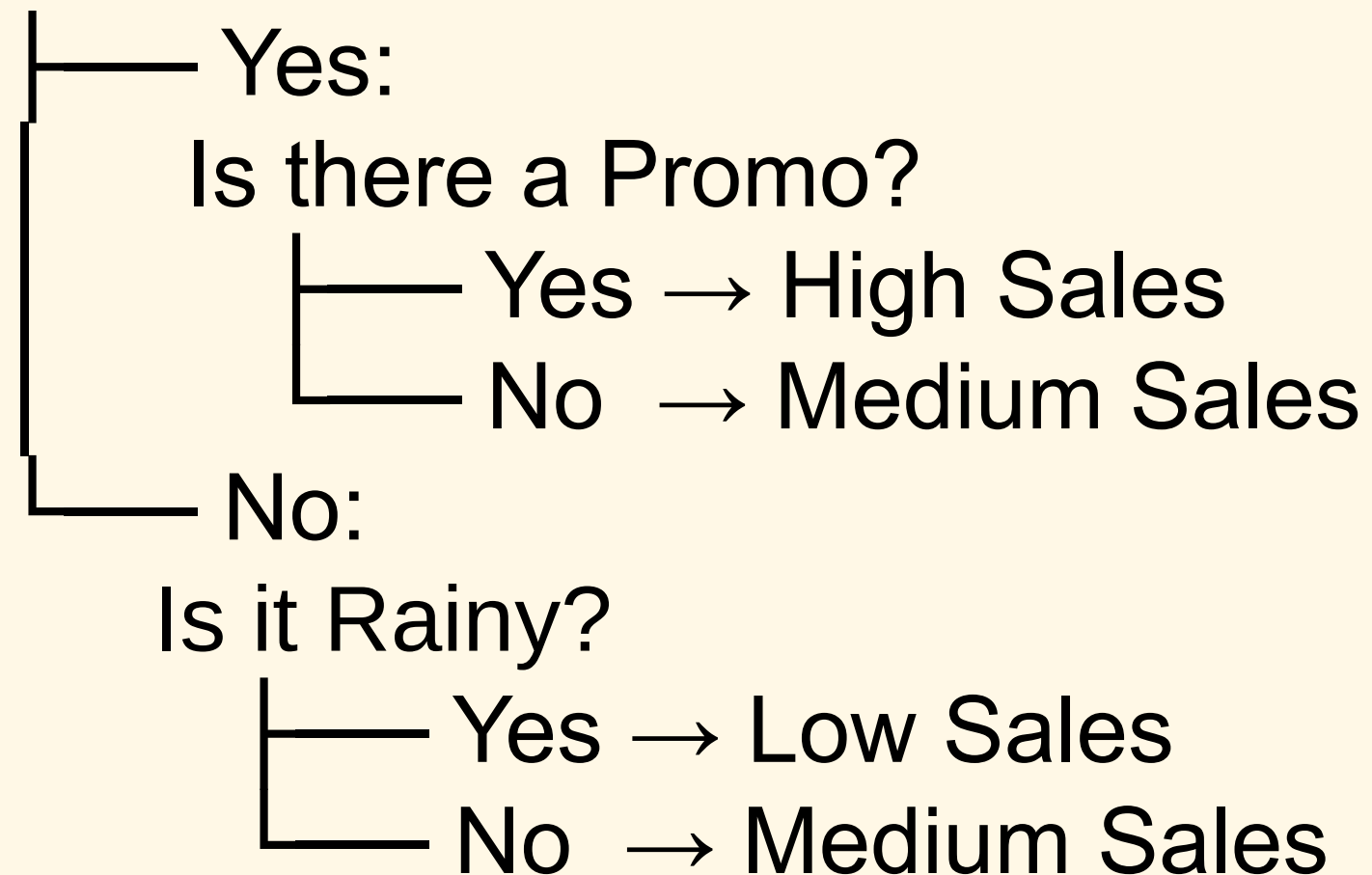
Example Tree



DECISION TREE–BASED METHODS

Decision Tree = Nested If–Else

Is it Weekend?



A Decision Tree is just a **visual + data-driven way of building nested if–else rules automatically**

DECISION TREE–BASED METHODS

Strengths

- ✓ Easy to interpret
- ✓ Works well with categorical data

Weaknesses

- ✗ Overfitting
- ✗ Sensitive to noise

RULE-BASED METHODS

Instead of trees, we use **IF–THEN** rules

Jollibee Example

IF Day = Weekend AND Promo = Yes → High Sales

IF Weather = Rainy AND Promo = No → Low Sales

How Rules Are Generated

- ✓ Derived from decision trees OR
- ✓ Learned directly (e.g., RIPPER, Apriori-based)

RULE-BASED METHODS - STRUCTURE OF A RULE

General Form

IF condition(s) → class label

Components:

1. Antecedent (IF part)

- Conditions on features
 - ✓ Day = Weekend, Promo = Yes

2. Consequent (THEN part)

- Predicted class
 - ✓ High Sales

IF Time = 12PM AND Day = Sunday → High Sales

Multiple Conditions (Conjunction)

IF Day = Weekend AND Weather = Sunny AND Promo = Yes → High Sales

- Uses **AND** logic

RULE-BASED METHODS - RULE SET (THE CLASSIFIER)

Example Rule Set

R1: IF Weekend AND Promo → High

R2: IF Rainy AND No Promo → Low

R3: IF Weekday AND Lunch Time → Medium

➤ Together, these rules form the classifier

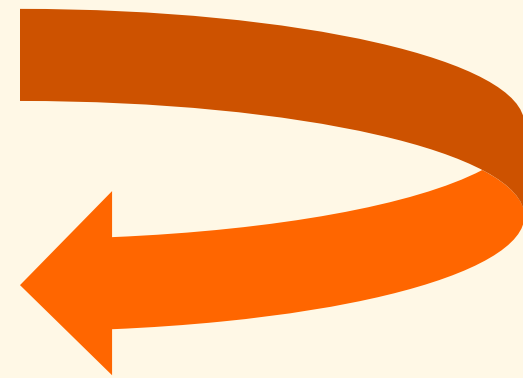
What if Multiple Rules Apply?

Example:

- ✓ Weekend + Rainy + Promo
- Could match:
- ✓ R1 and R2

Conflict Resolution Strategies

1. Rule Priority (Ordering)
2. Highest Confidence Rule
3. First Match Wins



RULE-BASED METHODS - HOW RULES ARE GENERATED

1. From Decision Trees

Convert paths into rules:

Tree path:

Weekend → Promo → High

Becomes:

IF Weekend AND Promo → High

2. Association Rule-Based Classification

From frequent patterns:

Example:

Promo → High Sales



How many rules can we derived from this?

RULE-BASED METHODS - TYPES OF RULE-BASED CLASSIFIERS

1. Ordered Rule Set

Rules are evaluated in sequence:

- ✓ $R1 \rightarrow R2 \rightarrow R3$
- ✓ First match is used

2. Unordered Rule Set

- ✓ All rules considered
- ✓ Use voting or scoring

RULE-BASED METHODS – ADVANTAGES and Limitations

Why Use Rule-Based Methods?

- ✓ Highly interpretable
- ✓ Easy to explain to managers
- ✓ Transparent decision-making

Perfect for business like Jollibee

Challenges

- ✗ Rule explosion (too many rules)
- ✗ Conflicting rules
- ✗ Hard to manage with many features

RULE-BASED METHODS – Jollibee Examples

1. Sales Prediction

- ✓ IF Weekend AND Promo → High Sales

2. Staffing Decisions

- ✓ IF High Sales → Add Staff

3. Inventory Planning

- ✓ IF Rainy → Increase Delivery Items

4. Customer Behavior

- ✓ IF Student AND Lunch Time → Budget Meals

RULE-BASED METHODS – COMPARISON WITH DECISION TREES

Aspect	Decision Tree	Rule-Based
Structure	Tree	Flat rules
Interpretation	Visual	Logical
Flexibility	Less	More

MEMORY-BASED REASONING (k-NN)

Memory-based reasoning (specifically **k-NN**) is a method where:
We classify a new case based on the most similar past cases

Key Principle

“Similar situations → similar outcomes”

At Jollibee:

Instead of building rules like:

- “Weekend → High Sales”

- ✓ We ask:

“What happened on days most similar to today?”

MEMORY-BASED REASONING (k-NN) - WHY IT IS CALLED “MEMORY-BASED”

Reason

- ✓ It **stores all past data**
 - ✓ No explicit model (no equations, no rules)
- It “remembers” everything

Contrast:

Method	Learns Model?
Decision Tree	Yes
Naïve Bayes	Yes
k-NN	✗ No

MEMORY-BASED REASONING (k-NN) - COMPUTATIONAL EXAMPLE (JOLLIBEE)

New Case

Predict sales for:

Time = 5 PM

Promo = Yes (encode Yes=1, No=0)

Training Data

Time	Promo	Sales
4	1	High
6	1	High
2	0	Low
7	1	High
3	0	Low

STEP 1: UNDERSTAND THE FEATURES

We are using:

- ✓ **Time** (numeric)
- ✓ **Promo** (Yes=1, No=0)
- So each row is a point: (4,1), (6,1), (2,0), (7,1), (3,0)

STEP 2: COMPUTE DISTANCE

We use **Euclidean Distance**:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Compare each training point with **(5,1)**

MEMORY-BASED REASONING (k-NN) - COMPUTATIONAL EXAMPLE (JOLLIBEE)

Training Data

Time	Promo	Sales
4	1	High
6	1	High
2	0	Low
7	1	High
3	0	Low

Distance Calculations

1. (4,1) → High

$$d = \sqrt{(5 - 4)^2 + (1 - 1)^2} = \sqrt{1 + 0} = 1$$

2. (6,1) → High

$$d = \sqrt{(5 - 6)^2 + (1 - 1)^2} = \sqrt{1 + 0} = 1$$

3. (2,0) → Low

$$d = \sqrt{(5 - 2)^2 + (1 - 0)^2} = \sqrt{9 + 1} = \sqrt{10} \approx 3.16$$

4. (7,1) → High

$$d = \sqrt{(5 - 7)^2 + (1 - 1)^2} = \sqrt{4} = 2$$

5. (3,0) → Low

$$d = \sqrt{(5 - 3)^2 + (1 - 0)^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.24$$

MEMORY-BASED REASONING (k-NN) - COMPUTATIONAL EXAMPLE (JOLLIBEE)

STEP 3: SORT DISTANCES

Point	Sales	Distance
(4,1)	High	1
(6,1)	High	1
(7,1)	High	2
(3,0)	Low	2.24
(2,0)	Low	3.16

STEP 4: CHOOSE k

Let's use: **k = 3**

So we pick the **3 nearest neighbors**

STEP 5: SELECT NEAREST NEIGHBORS, Top 3 closest:

Point	Sales
(4,1)	High
(6,1)	High
(7,1)	High

STEP 6: MAJORITY VOTING

Count the classes:

- ✓ High = 3
- ✓ Low = 0

FINAL ANSWER

Predicted Sales = **HIGH**

At Jollibee:

“The most similar past situations all had HIGH sales, so this new situation will likely also have HIGH sales.”

WHAT IF k CHANGES?

If **k = 5**:

Neighbors:

- ✓ High, High, High, Low, Low
- ✓ High = 3, Low = 2
Still **High**, but less confident

NEURAL NETWORKS / DEEP LEARNING

A Neural Network is a model inspired by the human brain that:

- ✓ Learns patterns by adjusting **weights** between connected “neurons”
- ✓ Instead of writing rules, the model **learns complex relationships automatically**

At Jollibee:

Instead of:

“IF Weekend → High Sales” (rules)

Neural network learns:

“Weekend + Rain + Promo + Time → complex pattern → Sales”

NEURAL NETWORKS / DEEP LEARNING – Basic Structure

Layers

- ✓ **Input Layer** → features
- ✓ **Hidden Layer(s)** → processing
- ✓ **Output Layer** → prediction

Example (Jollibee)

Inputs:

- ✓ Time
- ✓ Day
- ✓ Weather
- ✓ Promo

Output:

- ✓ Sales Level (High / Low)

Conceptual Diagram



NEURAL NETWORKS / DEEP LEARNING – HOW A NEURON WORKS

Step 1: Weighted Sum

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Where:

- x = inputs
- w = weights
- b = bias

Step 2: Activation Function

$$\text{Output} = \sigma(z)$$

Common:

- Sigmoid
- ReLU

NEURAL NETWORKS / DEEP LEARNING – HOW A NEURON WORKS

Problem

Predict if sales = **High (1)** or **Low (0)**

Inputs (encoded)

Feature	Value
Time (scaled)	0.8
Promo	1
Weekend	1

Given Weights

- $w_1 = 0.5$
- $w_2 = 1.0$
- $w_3 = 0.7$
- Bias $b = -1$

Step 3: Decision

- If $\geq 0.5 \rightarrow$ High
- Else $\rightarrow$ Low

Prediction = **High Sales**

Step 1: Compute z $z = w_1x_1 + w_2x_2 + w_3x_3 + b$

$$z = (0.5)(0.8) + (1.0)(1) + (0.7)(1) - 1$$

$$z = 0.4 + 1 + 0.7 - 1 = 1.1$$

Step 2: Apply Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Substitute $z = 1.1$

$$\sigma(1.1) = \frac{1}{1 + e^{-1.1}}$$

Compute the exponential

We evaluate:

$$e^{-1.1}$$

Using a calculator:

$$e^{-1.1} \approx 0.3329$$

Add 1

$$1 + 0.3329 = 1.3329$$

Final Answer

$$\sigma(1.1) \approx 0.75$$



NEURAL NETWORKS / DEEP LEARNING – HOW THE MODEL LEARNS (TRAINING)

Process

1. Predict output
2. Compare with actual value
3. Compute error
4. Adjust weights

Backpropagation

- Updates weights using error
- Uses gradient descent

Analogy

Like a manager at Jollibee learning:

“Last time I predicted LOW but sales were HIGH → adjust thinking”

NEURAL NETWORKS / DEEP LEARNING – Advantages and disadvantages

ADVANTAGES

- ✓ Handles complex patterns
- ✓ Works with large data
- ✓ High accuracy

DISADVANTAGES

- ✗ Black box (hard to interpret)
- ✗ Needs large data
- ✗ Computationally expensive

NAÏVE BAYES & BAYESIAN BELIEF NETWORKS

Naïve Bayes is a classification technique based on **probability** and **Bayes' Theorem**:

- ✓ It predicts the class with the **highest probability given the observed features**
- ✓ “Given the evidence, which outcome is most likely?”

At Jollibee:

Instead of rules or distances, we ask:

“What is the probability that sales are HIGH given that it is Weekend and there is a Promo?”

NAÏVE BAYES & BAYESIAN BELIEF NETWORKS

- BAYES' THEOREM

Formula

$$P(C | X) = \frac{P(X | C) \cdot P(C)}{P(X)}$$

Where:

$P(C | X)$ =Probability of class given data

$P(X | C)$ =Likelihood

$P(C)$ =Prior probability

$P(X)$ =Evidence

Simplification in Classification

We compare:

$$P(C | X) \propto P(X | C) \cdot P(C)$$

We ignore $P(X)$ because it is the same for all classes

NAÏVE BAYES & BAYESIAN BELIEF NETWORKS

- “NAÏVE” ASSUMPTION

Assumption

Features are **independent**

$$P(X | C) = P(x_1 | C) \cdot P(x_2 | C) \cdot \dots$$

Jollibee Example

Assumes:

- Promo effect is independent of Weather
- Time is independent of Day
- ✓ Not always true—but works surprisingly well

NAÏVE BAYES & BAYESIAN BELIEF NETWORKS

- Example

Problem

Predict:

➤ Sales = High or Low

Given:

Weekend = Yes

Promo = Yes

Training Data

Case	Weekend	Promo	Sales
1	Yes	Yes	High
2	Yes	No	High
3	No	Yes	Low
4	Yes	Yes	High
5	No	No	Low

“Based on past data, HIGH sales are much more likely when it's weekend and there is a promo.”

Step 1: Compute Priors

$$P(High) = 3/5 = 0.6$$

$$P(Low) = 2/5 = 0.4$$

Step 2: Compute Likelihoods

For HIGH:

$$P(Weekend = Yes | High) = 3/3 = 1$$

$$P(Promo = Yes | High) = 2/3 \approx 0.67$$

For LOW:

$$P(Weekend = Yes | Low) = 0/2 = 0$$

$$P(Promo = Yes | Low) = 1/2 = 0.5$$

Step 3: Compute Posterior (Proportional)

HIGH:

$$P(High | X) \propto 0.6 \times 1 \times 0.67 = 0.402$$

LOW:

$$P(Low | X) \propto 0.4 \times 0 \times 0.5 = 0$$

FINAL ANSWER

Prediction = **HIGH SALES**

BAYESIAN BELIEF NETWORKS (BBN)

A **graphical model** that:

- ✓ Represents variables as nodes
- ✓ Shows dependencies using edges

Example (Jollibee)

Weather → Customer Volume → Sales

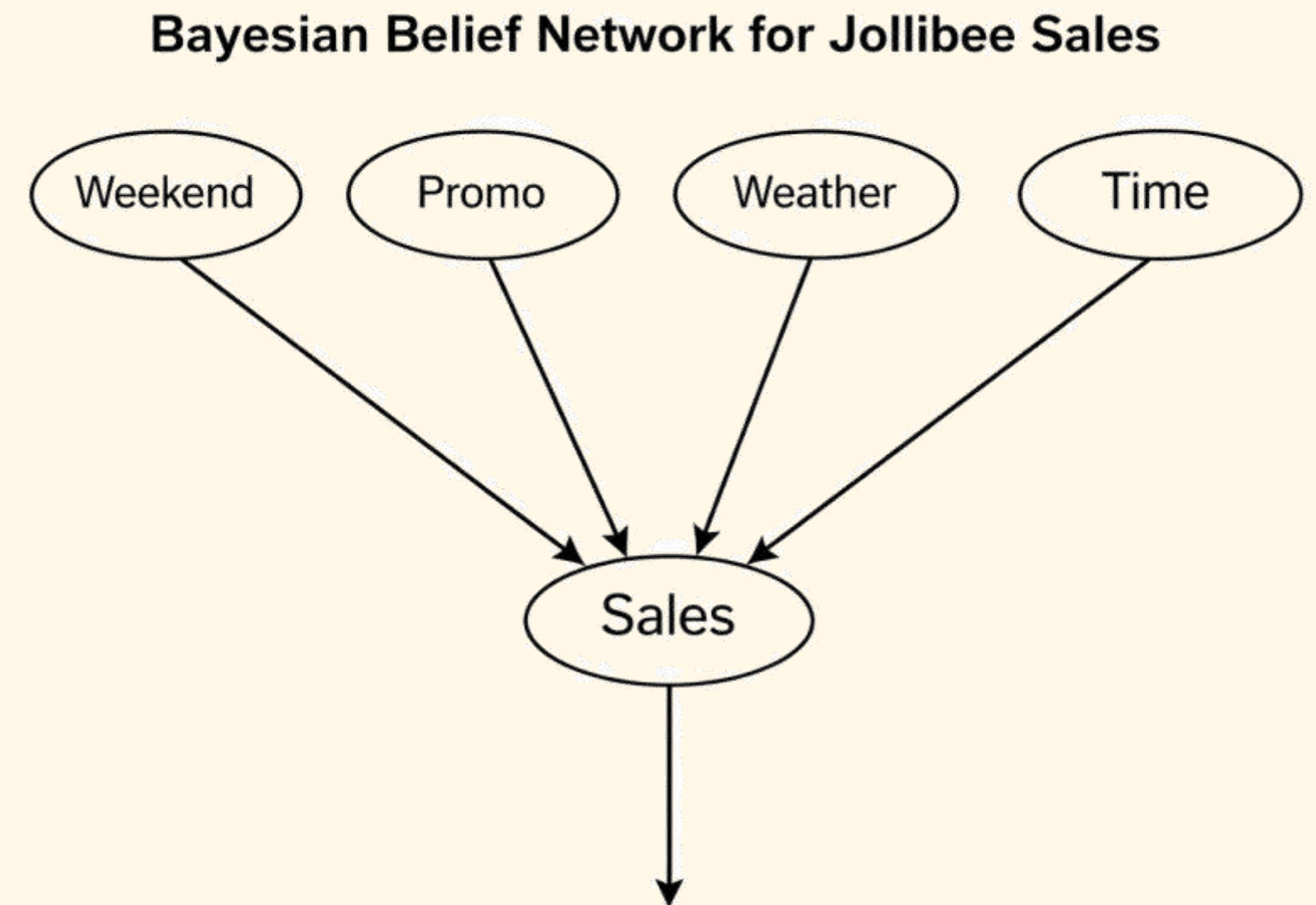
Promo → Customer Volume

Key Difference from Naïve Bayes

Naïve Bayes	Bayesian Network
Assumes independence	Models' dependencies
Simple	More complex
Faster	More expressive

BBN captures real-world relationships better

Naïve Bayes *is actually a special case* of a Bayesian Belief Network (BBN)



NAÏVE BAYES & BAYESIAN BELIEF NETWORKS

Naïve Bayes = “Simplified Thinking”

At Jollibee:

“Each factor affects sales separately.”

Like saying:

“Your exam score depends independently on study time, sleep, and attendance.”

BBN = “Real-World Thinking”

“Some factors influence each other before affecting sales.”

Like saying:

“Sleep affects concentration, which affects study effectiveness, which affects score.”

“Naïve Bayes is a simple Bayesian network where all features independently point to the target variable.”

NAÏVE BAYES & BAYESIAN BELIEF NETWORKS – Advantages & Disadvantages

ADVANTAGES

- ✓ Simple and fast
- ✓ Works well with small data
- ✓ Handles categorical data well
- ✓ Strong baseline model

DISADVANTAGES

- ✗ Independence assumption unrealistic
- ✗ Sensitive to probability estimation
- ✗ Zero probability issue

Naïve Bayes does not find patterns—it calculates **likelihoods**

At Jollibee:

- ✓ Decisions often involve uncertainty
- ✓ Probability is more realistic than strict rules

SUPPORT VECTOR MACHINES (SVM)

A Support Vector Machine (SVM) is a classification method that:

- ✓ Finds the **best boundary (hyperplane)** that separates classes while maximizing the margin between them
- ✓ “Don’t just separate classes—separate them as far apart as possible.”

At Jollibee:

We want to separate:

- ✓ High Sales
- ✓ Low Sales

Based on:

- ✓ Time
- ✓ Promo

SVM finds the **cleanest and safest dividing line**

SUPPORT VECTOR MACHINES (SVM) - BASIC CONCEPTS

1. Hyperplane

A line (in 2D) or surface (in higher dimensions):

$$w \cdot x + b = 0$$

This separates the classes

2. Margin

Distance between:

- ✓ The boundary (hyperplane)
- ✓ The closest data points

3. Support Vectors

- ✓ The **closest points to the boundary**

These are the most important points

Key Insight

Only the **support vectors** determine the boundary

SUPPORT VECTOR MACHINES (SVM) - VISUAL INTUITION

Imagine plotting:

X-axis → Time

Y-axis → Promo

Points: High Sales → ● , Low Sales → ●

Many lines can separate them

SVM chooses the line with **maximum margin**

WHY IS THAT THE BEST?

Even though you see multiple possible lines (dashed lines), SVM chooses the one that:

1. Separates the classes correctly

- ✓ All ● (High Sales) are on one side
- ✓ All ● (Low Sales) are on the other

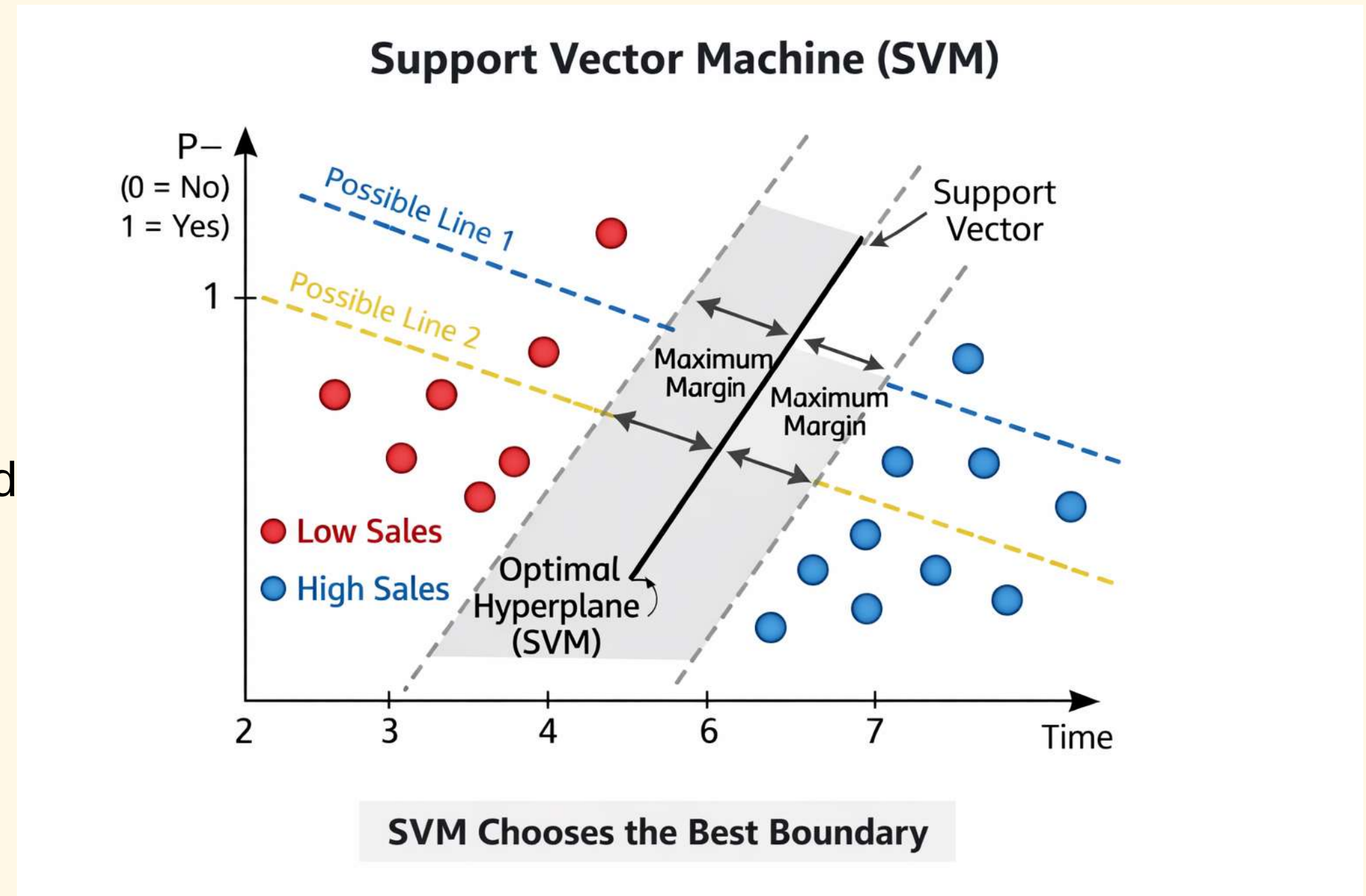
2. Maximizes the margin (MOST IMPORTANT)

The margin = distance between:

- ✓ The boundary (line)
- ✓ The closest data points (support vectors)

In the image:

- ✓ The **gray parallel lines** show the margin
- ✓ The **black line is exactly in the middle**



SUPPORT VECTOR MACHINES (SVM) – Advantages & Disadvantages

ADVANTAGES

- ✓ Works well with high-dimensional data
- ✓ Effective even with small datasets
- ✓ Robust to overfitting

DISADVANTAGES

- ✗ Hard to interpret
- ✗ Sensitive to parameter tuning
- ✗ Computationally expensive

SUMMARY TABLE: CLASSIFICATION TECHNIQUES

Technique	Core Idea	How It Works	Jollibee Example	Strengths	Weaknesses	Best Use Case
Decision Tree	Nested if–else rules	Splits data into branches using conditions	“If Weekend & Promo → High Sales”	✓ Easy to interpret✓ Visual✓ Handles categorical data	✗ Overfitting✗ Sensitive to noise	When explainability is important
Rule-Based Methods	IF–THEN rules	Uses a set of logical rules	“IF Rainy & No Promo → Low Sales”	✓ Human-readable✓ Transparent decisions	✗ Rule conflicts✗ Too many rules	Policy-based systems, business rules
Memory-Based (k-NN)	Similar cases → similar outcomes	Finds nearest neighbors using distance	“Similar past days had High Sales”	✓ Simple✓ No training phase	✗ Slow✗ Needs scaling✗ Sensitive to noise	Small datasets, similarity problems
Neural Networks / Deep Learning	Weighted neurons learn patterns	Layers compute weighted sums + activation	Learns complex patterns (e.g., rain affects delivery vs dine-in)	✓ High accuracy✓ Handles complex relationships	✗ Black box✗ Needs large data	Complex, large-scale problems
Naïve Bayes	Probability-based classification	Uses Bayes' Theorem with independence assumption	“Probability of High Sales given Weekend + Promo”	✓ Fast✓ Works well with small data	✗ Independence assumption unrealistic✗ Zero probability issue	Text classification, quick predictions
Support Vector Machine (SVM)	Optimal separating boundary	Finds hyperplane with maximum margin	Separates High vs Low Sales with best boundary	✓ High accuracy✓ Works in high dimensions	✗ Hard to interpret✗ Computationally heavy	Clear boundary problems, medium datasets

SUMMARY TABLE: key dimensions

Aspect	Decision Tree	Rule-Based	k-NN	Neural Net	Naïve Bayes	SVM
Interpretability	★ ★ ★ ★	★ ★ ★ ★	★ ★	★	★ ★ ★	★
Training Speed	Medium	Medium	None	Slow	Fast	Medium
Prediction Speed	Fast	Fast	Slow	Fast	Fast	Medium
Handles Non-linearity	Limited	Limited	Yes	Yes	Limited	Yes
Data Size Suitability	Medium	Medium	Small	Large	Small–Medium	Medium

SUMMARY

“Classification means deciding a category using data, and different algorithms solve this problem in different ways.”

Technique	How It Thinks
Decision Tree	Asks step-by-step questions
Rule-Based	Follows logical rules
k-NN	Looks at similar past cases
Naïve Bayes	Computes probabilities
Neural Network	Learns hidden patterns
SVM	Draws the best separating boundary

All techniques aim to solve the same problem—but **use different perspectives**

INTEGRATED JOLLIBEE SCENARIO

Scenario: Smart Branch Decision System

A branch of Jollibee wants to build a system that answers:
“Will today’s sales be HIGH or LOW?”

Given Inputs:

- Day (Weekend / Weekday)
- Weather (Sunny / Rainy)
- Promo (Yes / No)
- Time

1. Decision Tree

Approach:

Ask questions step-by-step

Example:

Is it Weekend?

→ Yes → Is there a Promo? → High

→ No → Low

Thinking style: Structured decision-making

2. Rule-Based

Approach:

Apply rules

IF Weekend AND Promo → High

IF Rainy AND No Promo → Low

Thinking style: Policy-based reasoning

3. k-NN (Memory-Based)

Approach:

Look at similar past days

“Days like today had HIGH sales”

Thinking style: Experience-based reasoning

INTEGRATED JOLLIBEE SCENARIO

4. Naïve Bayes

Approach:

Compute probability

“There is a 75% chance sales will be HIGH”

Thinking style: Probabilistic reasoning

The answer may be the same, but the reasoning is different

5. Neural Networks

Approach:

Learn hidden patterns

“Combination of time + weather + promo → complex pattern”

Thinking style: Pattern recognition

6. SVM

Approach:

Find best dividing line

“This combination falls in HIGH region”

Thinking style: Boundary optimization

Understanding Student Feedback through Multi-Class Text Classification.

Background

The Philippine government implemented the **Free Tuition Fee Act** to support students in state universities. To evaluate its effectiveness, students who are beneficiaries were asked:

“What is your overall feedback on the Free Tuition Fee Act program?”

The responses vary—some students express gratitude, others mention family benefits, motivation, or even challenges.

As data analysts, your task is to **manually analyze and classify these responses** using the concept of **multi-class classification**.

Learning Objectives

At the end of this activity, students should be able to:

1. Understand **classification in data mining**
2. Differentiate **binary, multi-class, and multi-label classification**
3. Apply **manual classification techniques** to real-world text data
4. Explain how different **classification algorithms** would approach the problem

Part A: Understanding the Problem (Guide Questions)

Answer briefly:

1. Why is this problem considered a **classification problem**?
2. Why is this specifically a **multi-class classification** task?
3. Can a single response belong to more than one category? If yes, what type of classification is that?

Part B: Define the Classes

You are given the following categories:

- ✓ Appreciation
- ✓ Family-Oriented Impact
- ✓ Academic Motivation
- ✓ Challenges / Issues
- ✓ Neutral / General Feedback

Tasks:

1. Explain each category in your own words
2. Provide 1 example sentence for each category

Part C: Manual Classification Activity

Below are actual student responses. Classify each response into **ONE best**

Ca	No.	Student Response	Your Class
	1	“I am very thankful because I can continue my studies.”	
	2	“This program really helped my parents financially.”	
	3	“It motivates me to study harder and not waste the opportunity.”	
	4	“Sometimes the process is slow and confusing.”	
	5	“The program is okay and helpful.”	
	6	“Because of this, my family has less financial burden.”	
	7	“I feel more inspired to finish my degree.”	
	8	“There are delays in the release of benefits.”	
	9	“I appreciate the government for this support.”	
	10	“It is a good program but needs improvement.”	

Part D: Analysis and Reflection

Answer the following:

1. Which responses were difficult to classify? Why?
2. Did you encounter responses that could belong to more than one category? Give an example.
3. What challenges arise when classifying text data manually?

Part E: Connecting to Classification Techniques

Explain (in simple terms) how the following methods would classify the responses:

Decision Tree

What kind of “questions” would it ask?

Rule-Based Method

Give 2 example IF-THEN rules

Naïve Bayes

How would it use word probabilities?

K-Nearest Neighbors (KNN)

How does similarity between responses work?

Neural Networks

Why are they useful for text classification?

Support Vector Machine (SVM)

How does it separate categories?

Part F: Extension Question

How can this analysis help improve the implementation of the Free Tuition Fee Act?

Part G: Google Colab Coding “Multi-Class Text Classification”

Thank You!

Any Questions?